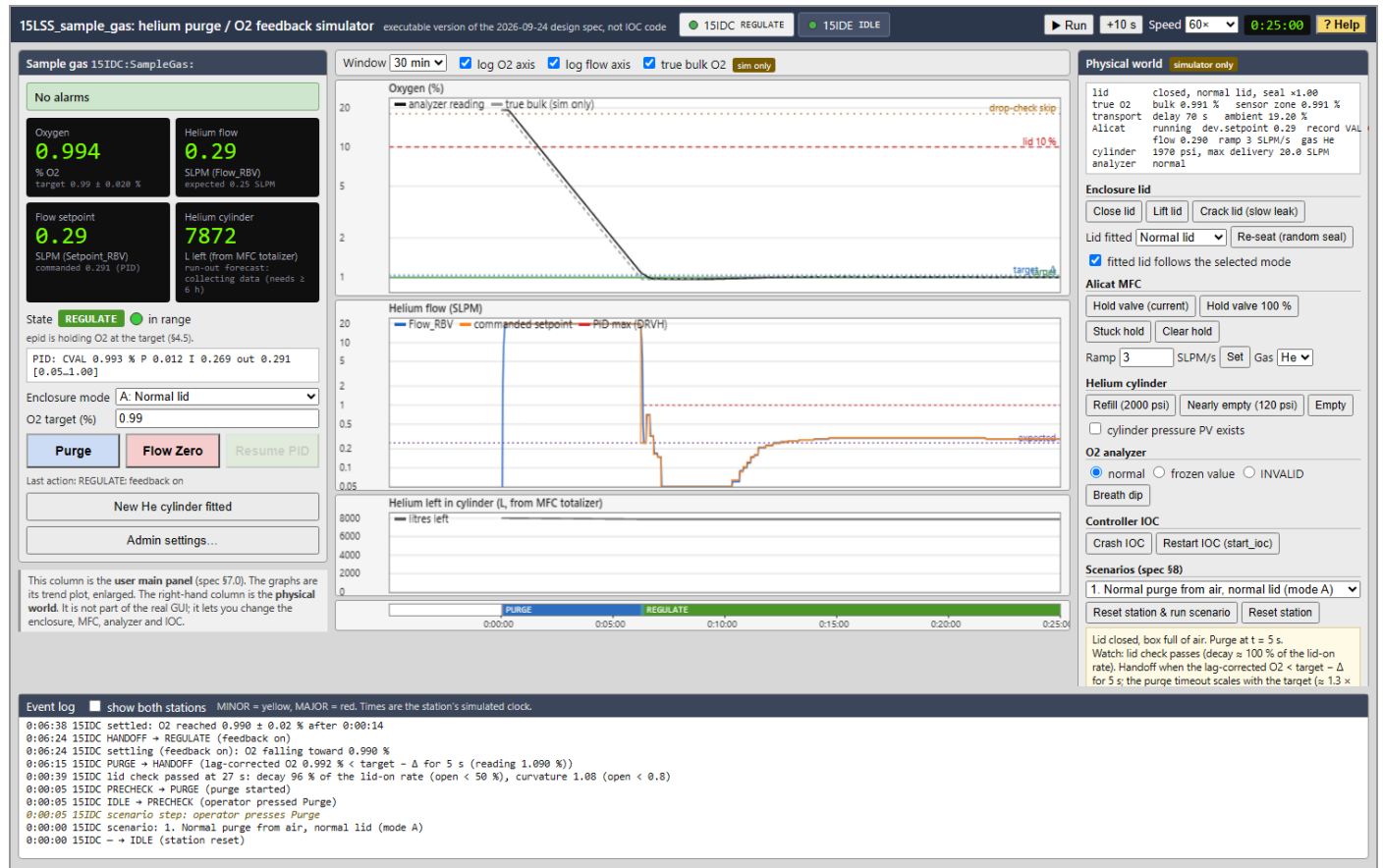


Sample-gas controller simulator: illustrated tour

This is a picture-by-picture companion to the text guide (`USER_GUIDE.md`, also under **? Help** in the simulator). Every image is generated from the simulator itself by `tools/make_screenshots.py`, so it can be refreshed after any change.

The URL in each caption reproduces the picture: open `simulator/sample_gas_simulator.html` followed by that #... text. Screenshot mode uses a fixed random seed, so the numbers quoted below reappear exactly.

1. The whole screen: regulating after a purge

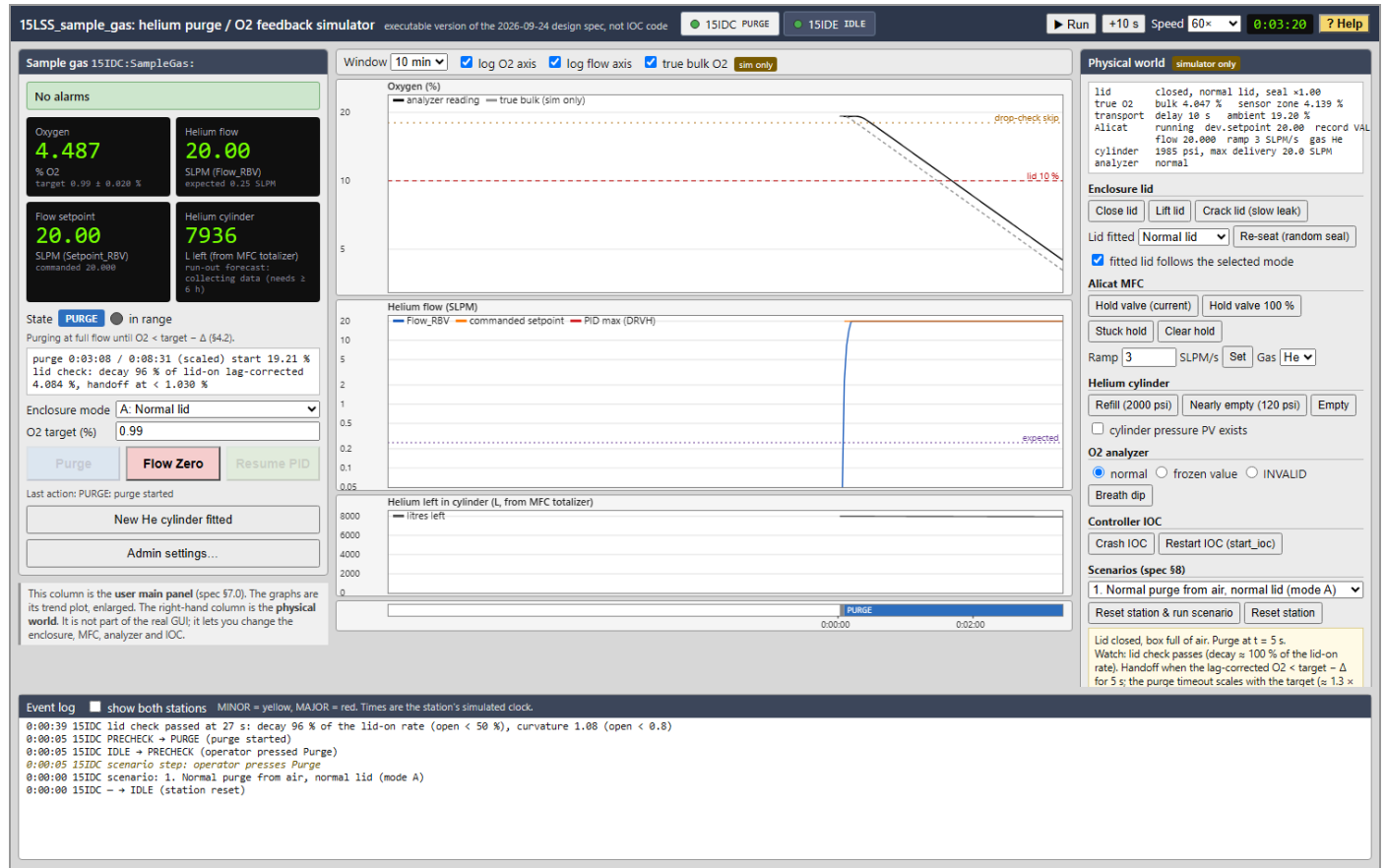


#shot=1&t=1500&win=1800: scenario 1, 25 min in.

- **Left column: the operator's main panel.**
 - green "No alarms" banner
 - the four readouts: O2 0.994 %, helium flow 0.29 SLPM, setpoint, helium left 7872 L
 - state **REGULATE** with the green in-range light
 - the PID line (CVAL, P, I, output and its limits)
 - mode, target and the buttons
- **Centre: the graphs.**
 - **O2 (log axis):** the purge from air down to 1 %. The black analyzer reading trails the grey true bulk O2. The green band is target $\pm$ tolerance.
 - **Flow:** the 20 SLPM purge, the step down at handoff, then the PID trimming around 0.25 SLPM.
 - **Helium left**, and the **state strip** (PURGE in blue, REGULATE in green).

- **Right column: the physical world** (simulator only): lid, Alicat, cylinder, analyzer, IOC and scenarios.
- **Bottom: the event log**, newest first. Read it bottom-up:
 - the lid check passed at 27 s (decay 96 % of the lid-on rate, curvature 1.08)
 - handoff when the lag-corrected O2 reached target – Δ
 - feedback on
 - "settled" 14 s later

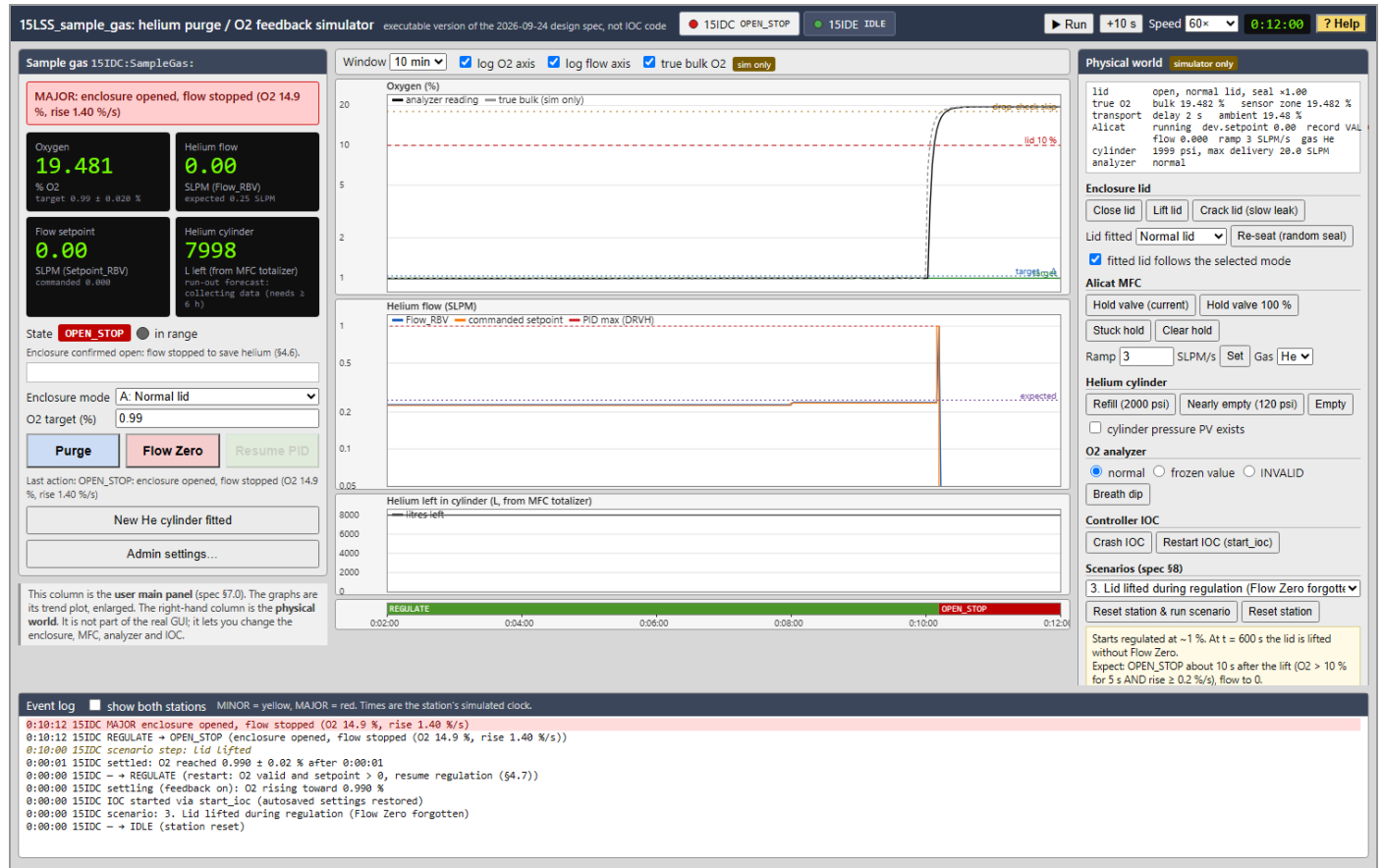
2. A purge in progress



#shot=1&t=200&win=600

- **Flow** is at 20 SLPM, and O2 is falling exponentially on the log axis.
- **The progress line shows:**
 - elapsed time against the purge timeout, which scales with the target
 - the lid-check result (decay 96 % of the lid-on rate, so the lid is on)
 - the **lag-corrected** O2 the handoff uses (about 10 % below the reading at full flow)
 - the handoff level (target – Δ ; $\Delta = -0.04$ %, so 1.03 %)
- **Purge** is greyed out while purging. **Flow Zero** stays available.

3. Lid lifted without Flow Zero: helium stopped



#shot=3&t=720&win=600: scenario 3, lid lifted at 10:00.

- **O2 jumps** from 1 % to ambient within seconds.
- **Twelve seconds after the lift**, the controller confirms "enclosure opened" (O2 > 10 % for 5 s **and** a fast rise) and drops the flow to **0**.
- **Red MAJOR banner, and state OPEN_STOP** (red in the state strip).
- **To recover:** close the lid, then press **Purge**.

4. Admin (beamline staff)

15LSS_sample_gas: helium purge / O2 feedback simulator

Sample gas 15LSS: SampleGas:

No alarms

Oxygen: 0.994

Helium flow: 0.29

Flow setpoint: 0.29

Helium cylinder: 7872

State: REGULATE in range

enclosure mode: A: Normal lid

O2 target (%): 0.99

Purge

Flow Zero

Admin settings...

Helium usage report (window ends at the end of the latest user run)

Live internals (read-only)

Setting	Value	Unit	Limits
O2 in-range tolerance (display only, not a deadband)	0.02	%	0.005 ... 0.5
Purge handoff margin Δ below target (negative = above target)	-0.04	%	-1 ... 1
Purge flow	20	SLPM	1 ... 20
Purge timeout = this $\times$ kinetic time to target - Δ	1.3	$\times$	1 ... 3
Cylinder forecast MINOR below	24	h	1 ... 240
Cylinder forecast MAJOR below	6	h	0.5 ... 120
Feedback delay after handoff (sensor setting)	0	s	0 ... 600
Flow-high MINOR threshold	1.5	$\times$ exp	1.05 ... 5
Flow-high MAJOR threshold	2	$\times$ exp	1.1 ... 10
PID pinned at max $\rightarrow$ MAJOR after	600	s	30 ... 7200
O2 "abnormally high" above target by	0.3	%	0.02 ... 5
Target not reached after feedback-on / target change within	7200	s	600 ... 86400

Setting	Mode A: Normal lid	Mode B: Collimator lid	Unit
epid KP at 0.99 % target (negative; scaled by 0.99/target)	-9.3	-10	SLPM/%
epid KI	0.00088	0.0014	1/s
Max PID flow (DRVH, capped by hard ceiling)	1	2	SLPM
Min PID flow (DRVH)	0.05	0.3	SLPM

Overrides (automatic corrections): count = 0

Helium usage report (window ends at the end of the latest user run)

Live internals (read-only)

#shot=1&t=1500&panel=admin

- Tolerance, Δ , purge settings, feedback delay, alarm thresholds, settle timeout and cylinder alarm levels.
- The **PID gains and flow limits per lid**, with KP stated at 0.99 % and scaled automatically.
- **Hover over any field** for its meaning, unit and allowed range. Out-of-range entries are clamped and the field flashes.
- **Further down:** the override log, the helium usage report and the live internals.
- **Deep admin settings...** is at the bottom.

5. Deep admin (instrument scientist)

15LSS_sample_gas: helium purge / O2 feedback simulator

Sample gas 15IDC:SampleGas:

No alarms

Oxygen 0.994 % O2 target: 0.99 ± 0.028 %

Helium flow 0.29 SLPM (Flow_RBV) expected 0.25 SLPM

Flow setpoint 0.29 SLPM (Setpoint_RBV) commanded 0.291 (PID)

Helium cylinder 7872 L left (from MFC total run-out forecast: collecting data (new 8 h))

State REGULATE in range

epid is holding O2 at the target (34.5%).

PID: CVAL 0.993 % P 0.012 I 0.269 out: 0.29 [0.05-1.00]

Enclosure mode A: Normal lid

O2 target (%) 0.99

Purge Flow Zero Resume

Last action: REGULATE: feedback on

New He cylinder fitted

Admin settings...

This column is the user main panel (spec \$7.0). The graph is its trend plot, enlarged. The right-hand column is the physical world. It is not part of the real GUI; it lets you change it: enclosure, MFC, analyzer and IOC.

Event log show both stations MINOR = yellow

0:06:38 15IDC settled: O2 reached 0.990 ± 0.028 %

0:06:24 15IDC HANDOFF → REGULATE (feedback on)

0:06:24 15IDC settling (feedback on): O2 fell

0:06:15 15IDC PURGE → HANDOFF (lag-corrected)

0:00:39 15IDC lid check passed at 27 s: decay

0:00:05 15IDC PRECHECK → PURGE (purge started)

0:00:05 15IDC IDLE → PRECHECK (operator pressed Purge)

0:00:05 15IDC scenario step: operator presses Purge

0:00:00 15IDC scenario: 1. Normal purge from air, normal lid (mode A)

0:00:00 15IDC → IDLE (station reset)

Deep admin: 15IDC:SampleGas: (instrument scientist)

Every physical and threshold number the controller uses. All are autosaved records with limits (DRV/L/DRV/H), so a typo cannot set an absurd value.

Enclosure modes

Setting	Mode A: Normal lid	Mode B: Collimator lid	Unit
Base flow for 0.99 % O2	0.25	0.84	SLPM
Expected-flow exponent n (flow ∝ (0.99/target)^n)	1	0.53	

Thresholds, timing and limits

Setting	Value	Unit	Limits
Enclosure volume (sets the lid-on purge decay rate F/V)	41	L	5 ... 200
Lid check: decay onset = drop of this fraction	0.02		0.005 ... 0.2
Lid check: no onset within this = open	30	s	10 ... 180
Lid check: decay measured over	15	s	4 ... 60
Lid check: open if decay levels off (2nd/1st-half rate) below	0.8		0.1 ... 1
Usable He in a full cylinder (run-out forecast)	8000	L	100 ... 50000
Helium usage report window	60	days	1 ... 365
User run closes after no purge for (Admin can also mark a new run)	216000	s	3600 ... 2592000
Lid check: open if decay < this fraction of F/V	0.5		0.05 ... 0.95
Lid check skipped if purge starts below	18	%	1 ... 21
Handoff: lag-corrected O2 below target - Δ for	5	s	1 ... 120
Handoff: analyzer lag at purge flow (0 = no compensation)	12	s	0 ... 60
Handoff: flow-settled tolerance	0.05	SLPM	0.01 ... 1
Lid detector: O2 level	10	%	2 ... 19
Lid detector: above level for	5	s	1 ... 60
Lid detector: rise rate (10 s difference)	0.2	%/s	0.005 ... 2
Lid detector: rate look-back window	60	s	10 ... 300
Lid detector arms in PURGE below	9	%	1 ... 18
Ramp clamp at purge start (max: 0 counts as instant)	5	SLPM/s	0.5 ... 20
Ramp clamp at handoff (min)	2	SLPM/s	0.1 ... 10
Hard PID flow ceiling (surface vibration)	2	SLPM	0.1 ... 20
Purge timeout: minimum	120	s	30 ... 1800

Physical world simulator only

closed, normal lid, seal <1.00

bulk 0.991 % sensor zone 0.991 %

delay 78 s ambient 19.28 %

running dev.setpoint 0.29 record VAL

flow 0.298 ramp 3 SLPM/s gas He

1970 psi, max delivery 20.0 SLPM

normal

Enclosure lid

Close lid Lift lid Crack lid (slow leak)

Fitted Normal lid Re-seat (random seal)

Fitted lid follows the selected mode

at MFC

Hold valve (current) Hold valve 100 %

Quick hold Clear hold

SLPM/s Set Gas He

Helium cylinder

Full (2000 psi) Nearly empty (120 psi) Empty

cylinder pressure PV exists

analyzer

normal frozen value INVALID

Leak dip

Controller IOC

Flash IOC Restart IOC (start_ioc)

Scenarios (spec \$8)

Normal purge from air, normal lid (mode A)

Reset station & run scenario Reset station

closed, box full of air. Purge at t = 5 s.

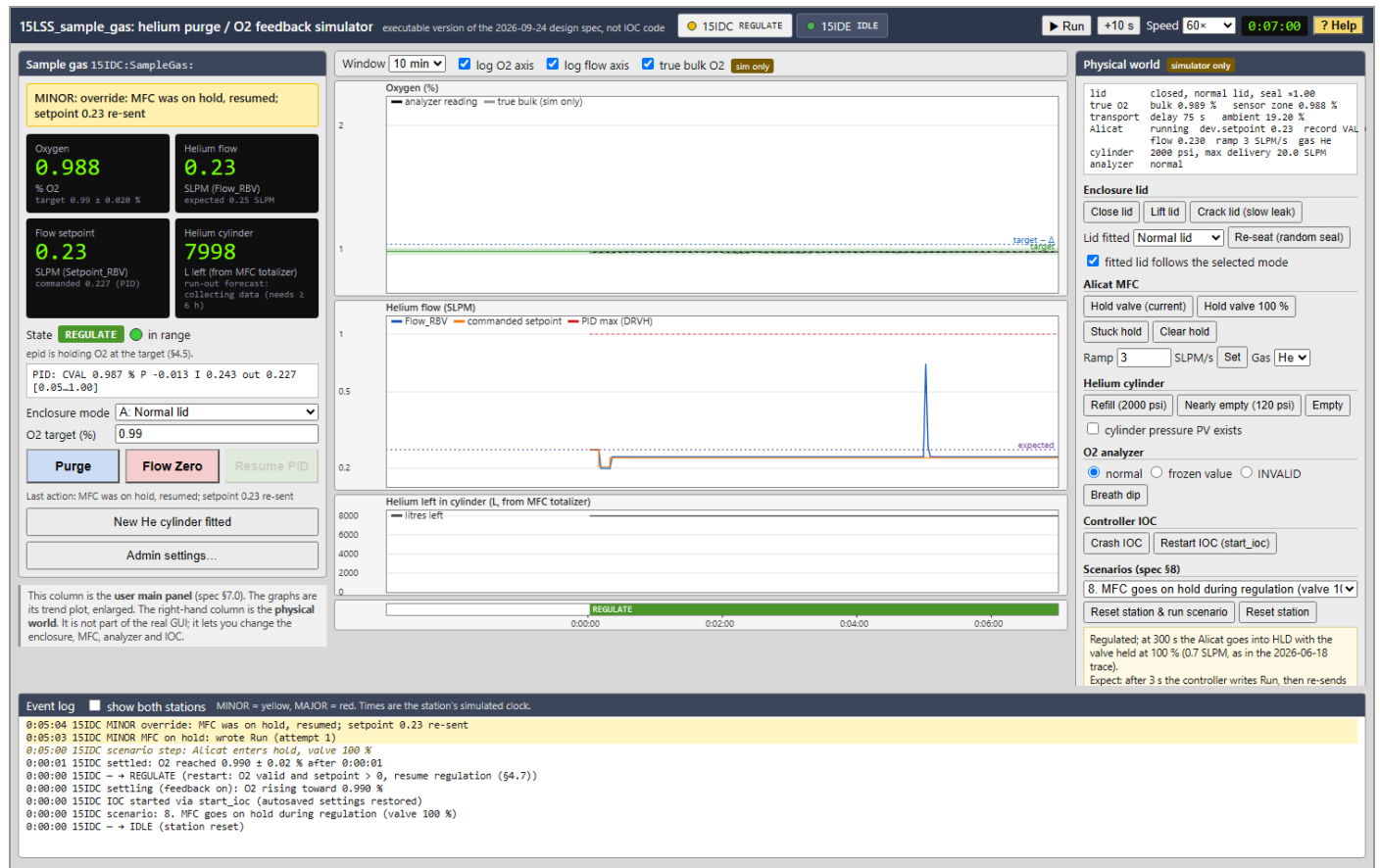
Match: lid check passes (decay ≈ 100 % of the lid-on rate). Handoff when the lag-corrected O2 < target - Δ

5 s; the purge timeout scales with the target (≈ 1.3 ×

#shot=1&t=1500&panel=deep

- Every physical and threshold number the controller uses:
 - enclosure modes (base flow, expected-flow exponent)
 - lid check (onset, window, rate and curvature thresholds)
 - cylinder capacity, report window, user-run gap
 - handoff lag correction, lid detector, ramp clamps, flow ceiling, purge-timeout clamps
 - mismatch, hold monitor, O2 validity, averaging, epid scan and deadband
 - gain scheduling, the near-target fine band, and the settling and slope logic
- In the real IOC each of these is an autosaved record with limits.

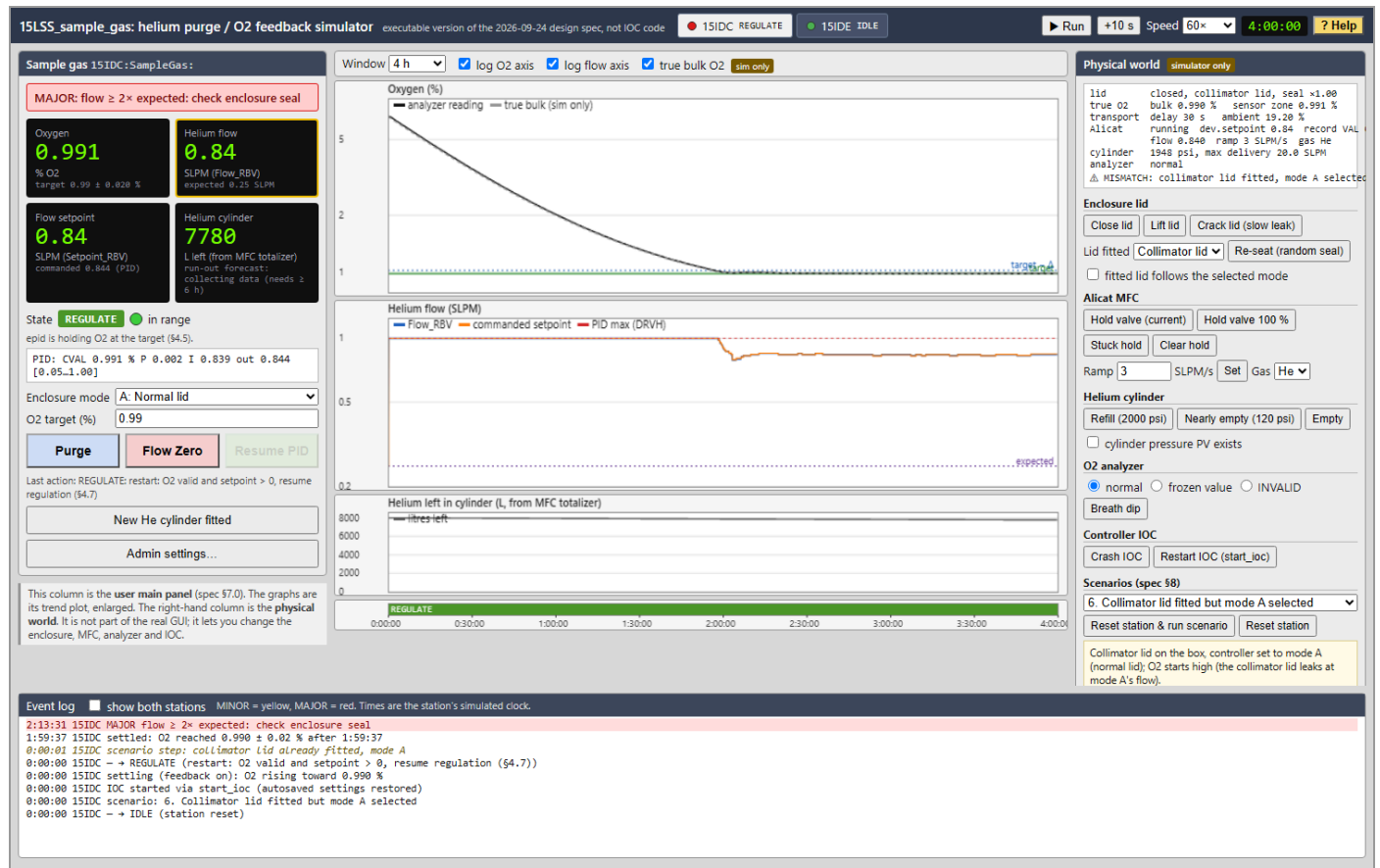
6. Alicat on hold: resumed automatically



#shot=8&t=420&win=600: scenario 8.

- At 5:00 the Alicat went into hold with the valve at 100 % (the world button "Hold valve 100 %", as seen in the June serial trace). The flow spikes.
- Three seconds later the controller wrote **Run** and re-sent its setpoint. The yellow MINOR banner says so, and the flow is back under PID control.
- The scenario description (bottom right) says what to watch for. **Reset station & run scenario** replays it.

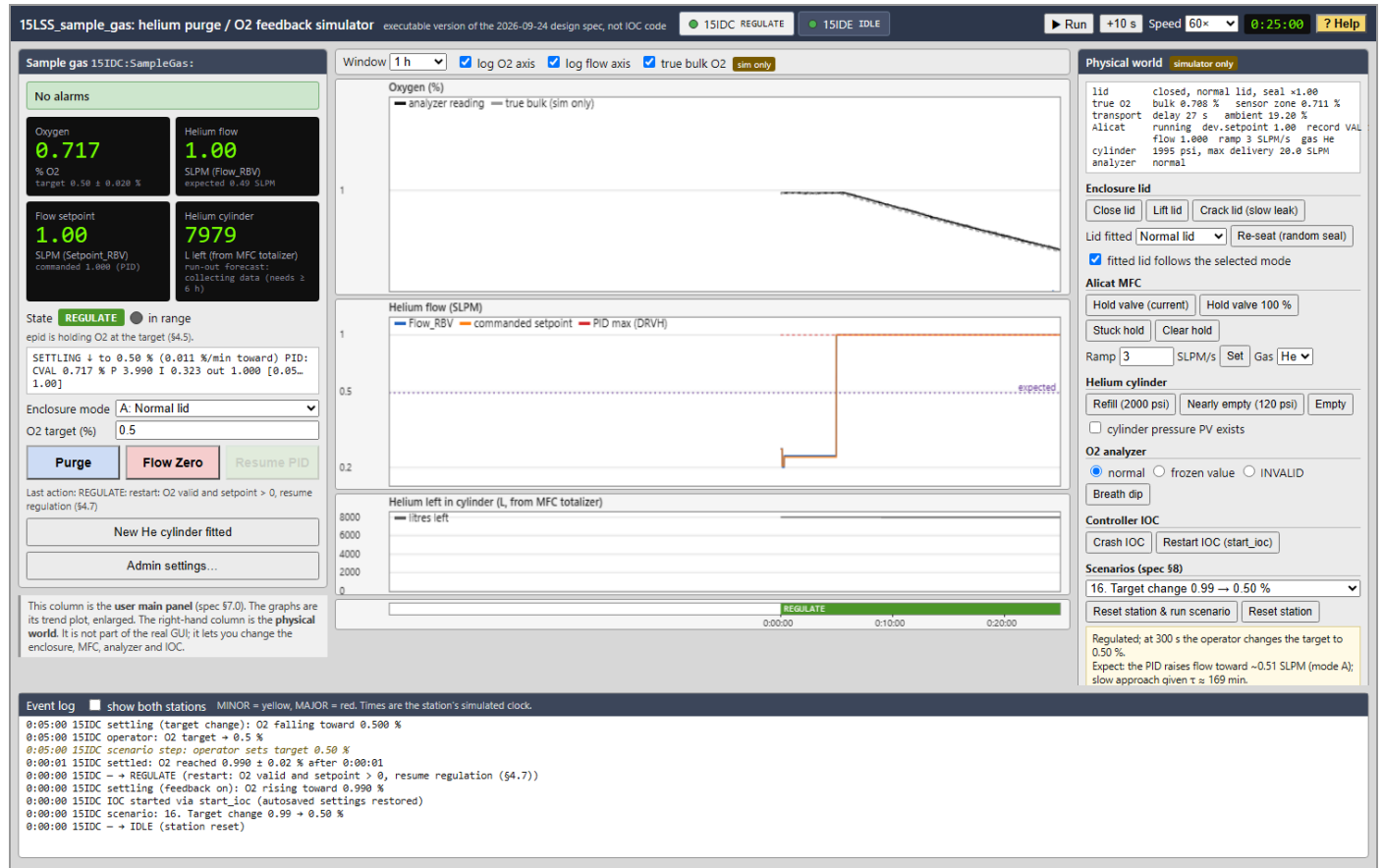
7. Wrong lid for the selected mode



#shot=6&t=14400&win=14400: scenario 6.

- **The box has the collimator lid but mode A (normal lid) is selected.** The world panel shows "⚠ MISMATCH", and "fitted lid follows the selected mode" is unticked, because this mismatch is deliberate.
- **The PID needs ~0.83 SLPM** instead of mode A's expected 0.25. While O2 was still falling toward target (settling), no flow alarm fired.
- **After O2 settled (~2 h),** the MAJOR alarm "flow $\geq 2\times$ expected: check enclosure seal" appeared. That is the right diagnosis: this enclosure needs far more helium than mode A assumes.

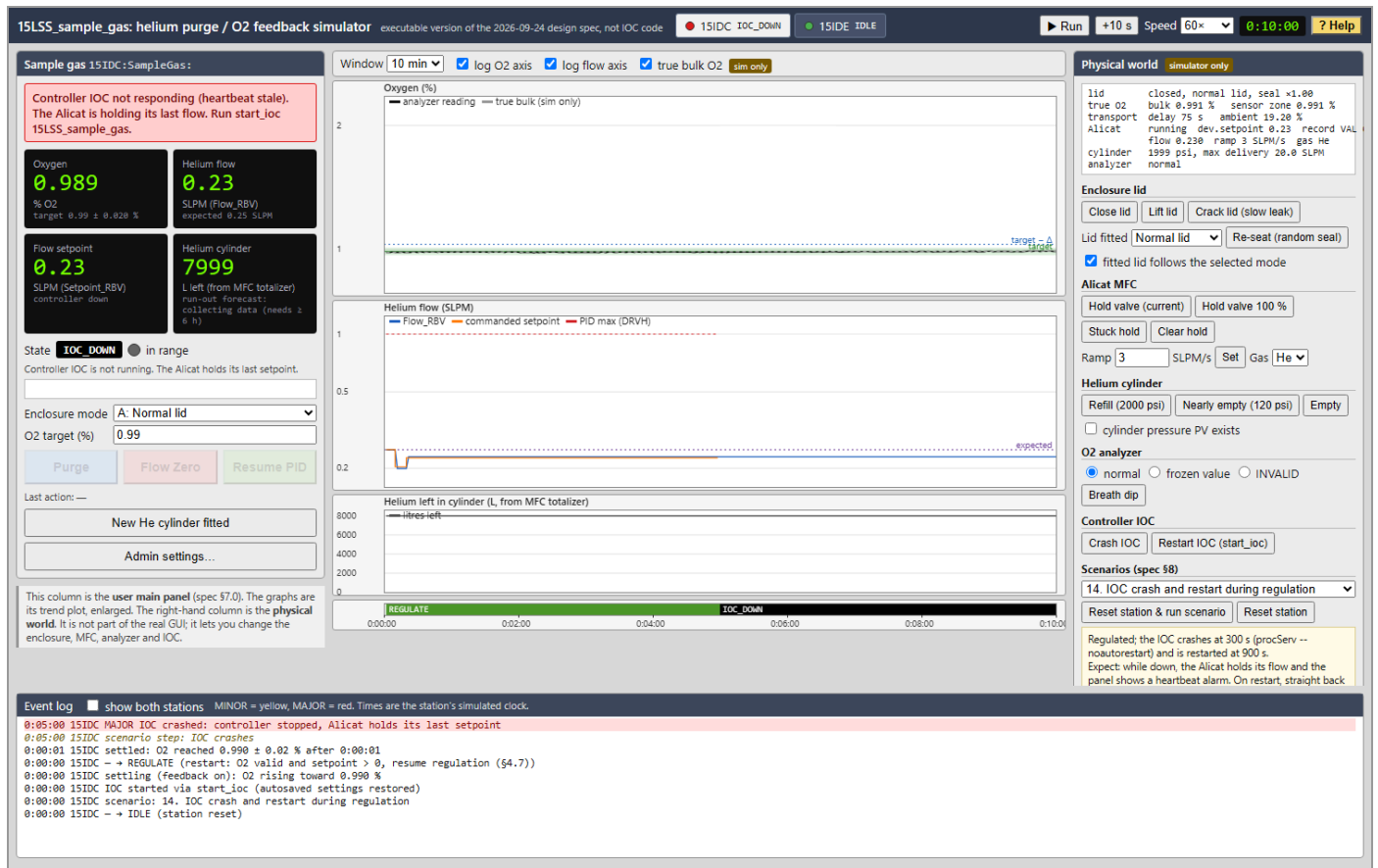
8. Changing the target: settling, no false alarms



#shot=16&t=1500&win=3600: scenario 16, target changed from 0.99 % to 0.50 % at 5:00.

- The progress line reads **"SETTLING ↓ to 0.50 %"** and shows how fast O2 is approaching the new target.
- The PID is at its **1.0 SLPM maximum**, but no alarm fires, because O2 keeps moving toward the target.
- If O2 stalled, or took more than 2 h, the flow and O2 alarms would apply at once.

9. Controller IOC down



#shot=14&t=600&win=600: scenario 14, IOC crashed at 5:00.

- **Banner:** "Controller IOC not responding (heartbeat stale)". The state is **IOC_DOWN**, and the buttons are disabled.
- **The Alicat keeps its last setpoint**, so the enclosure stays purged.
- **In the scenario**, the IOC is restarted at 15:00 and resumes regulation without a bump.

10. Cylinder run-out forecast



#shot=17&t=172800&win=0: scenario 17 after 2 days, whole history shown.

- **Six-hourly sample changes:** each Flow Zero / lid-off / purge cycle is a spike in O2 and flow.
- **Helium left** falls in steps at each purge.
- **The cylinder readout:** 6317 L left, "empty in 7.5–7.6 d (median 7.5 d, 4 windows)". The windows are the 3, 2, 1, 0.5 and 0.25-day usage fits that have enough data. Their spread is the confidence range, and early on it is wide.

11. Helium usage report

15LSS_sample_gas: helium purge / O2 feedback simulator

Sample gas 15IDC: SampleGas: No alarms

Oxygen 0.990 % O2 target 0.99 ± 0.028 %

Helium flow 0.25 SLPM (Flow_RBV) expected 0.25 SLPM

Flow setpoint 0.25 SLPM (Setpoint_RBV) commanded 0.255 (PID)

Helium cylinder 2805 L left (from MFC total: empty in 3.9 d, 4.5 d; median 4.7 d, 5 min)

State REGULATE in range

epid is holding O2 at the target (\$4.5).

PID: CVAL 0.990 % P 0.002 I 0.258 out 0.25 [0.05-1.00]

Enclosure mode A: Normal lid

O2 target (%) 0.99

Purge Flow Zero Resume

Last action: REGULATE: feedback on

New He cylinder fitted

Admin settings...

This column is the user main panel (spec 77.0). The graph is trend plot, enlarged. The right-hand column is the physical world. It is not part of the real GUI; it lets you change the enclosure, MFC, analyzer and IOC.

Event log show both stations MINOR = yellow

210:23:14 15IDC settled: O2 reached 0.990 ± 0.028 %

210:23:00 15IDC HANDOFF → REGULATE (feedback)

210:23:00 15IDC settling (feedback on): O2 feedback

210:22:51 15IDC PURGE → HANDOFF (lag-corrected)

210:17:14 15IDC lid check passed at 27 s: det

210:16:40 15IDC PRECHECK → PURGE (purge started)

210:16:40 15IDC FLOW_ZERO → PRECHECK (operator pressed Purge)

210:16:40 15IDC scenario step: Purge

210:16:00 15IDC scenario step: Lid closed

210:06:00 15IDC scenario step: Lid lifted

210:05:00 15IDC REGULATE → FLOW_ZERO (operator pressed Flow Zero)

Admin: 15IDC:SampleGas: (beamline staff)

O2 "abnormally high" above target by 0.3 % 0.02 ... 5

Target not reached after feedback-on / target change within 7200 s 600 ... 86400

PID gains and limits per enclosure mode

Setting	Mode A: Normal lid	Mode B: Collimator lid	Unit
epid KP at 0.99 % target (negative; scaled by 0.99/target)	-9.3	-10	SLPM/%
epid KI	0.00088	0.0014	1/s
Max PID flow (DRVH, capped by hard ceiling)	1	2	SLPM
Min PID flow (DRVH)	0.05	0.3	SLPM

Overrides (automatic corrections): count = 0 reset count

(none)

Helium usage report (window ends at the end of the latest user run) Mark start of new user run

last 60 days (day 0.00 → day 9.00):

helium dispensed 5195 L (5195 L during user runs)

cylinders fitted 1 cylinder-equivalents used 0.65 (at 8000 L each)

user runs (end → first flow-zero after the last purge; a run closes after 2.5 d with no purge, or at a "new user run" mark):

#1 day 0.01 → day 8.00 23 purges 4775 L

#2 day 8.51 → in progress 2 purges 420 L (open)

Live internals (read-only)

state REGULATE heartbeat 777600 down false

epid FBON 1 VAL 0.990 CVAL 0.9904 P 0.0020 I 0.2581 OVAL 0.255 DRVL 0.050 DRVH 1.000

expected flow 0.250 SLPM last commanded 0.255 SLPM

lid detector: above-level 0 s, max rise (60 s) 0.0004 %/s, armed-in-purge true

hold monitor: on-hold 0 s, attempts 0 O2 ok true frozen false

settling false direction -1 rate toward target -0.0003 %/min no-progress 0 s

cylinder: 2805 of 8000 L left (Total_RBV 5194.9 L, baseline 0.0 L)

3 d window: empty in 3.9 d (712 L/day)

2 d window: empty in 4.7 d (601 L/day)

1 d window: empty in 5.7 d (488 L/day)

0.5 d window: empty in 3.7 d (760 L/day)

0.25 d window: empty in 5.4 d (518 L/day)

Assumption to verify on the real IOC: epid writes OUTL every scan while FBON = 1, so an unchanged output is re-sent every scan. The hold monitor re-sends explicitly either way.

Deep admin settings... Release control (→ IDLE)

physical world simulator only

closed, normal lid, seal <1.00

bulk 0.991 % sensor zone 0.991 %

delay 75 s ambient 19.20 %

running dev.setpoint 0.25 record VAL

flow 0.250 ramp 3 SLPM/s gas He

778 psi, max delivery 20.0 SLPM

analyzer normal

enclosure lid

close lid Lift lid Crack lid (slow leak)

fitted Normal lid Re-seat (random seal)

fitted lid follows the selected mode

at MFC

old valve (current) Hold valve 100 %

lock hold Clear hold

mp 3 SLPM/s Set Gas He

Helium cylinder

refill (2000 psi) Nearly empty (120 psi) Empty

cylinder pressure PV exists

analyzer

normal frozen value INVALID

bath dip

controller IOC

flash IOC Restart IOC (start_ioc)

scenarios (spec 58)

Helium ledger: three users, a 48 h downtime

asset station & run scenario Reset station

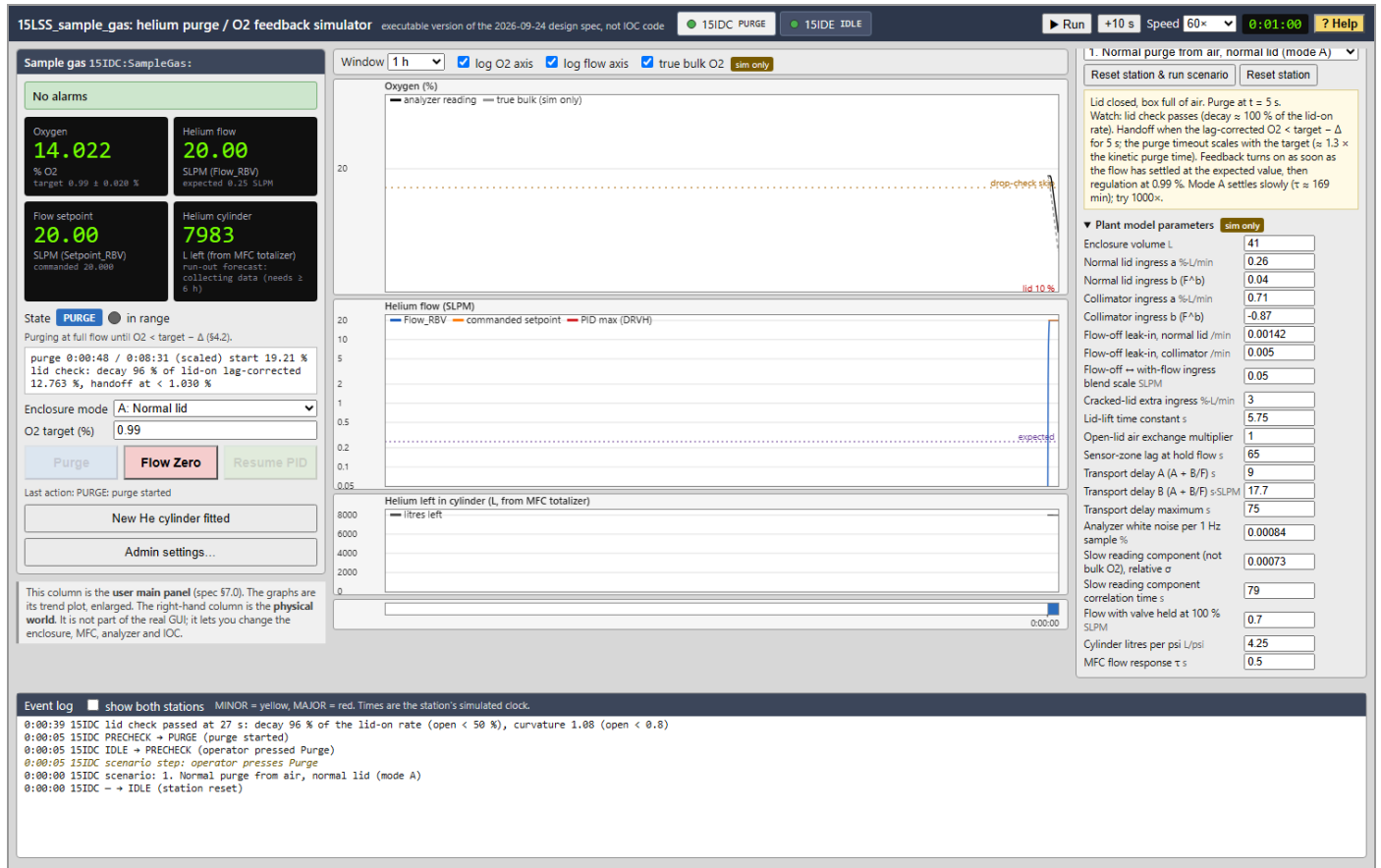
er A: days 0-3, then a 48 h beam downtime (flow at ro), resumes 10 h after the beam returns (day 5.4) until y 8 → ONE run.

changeover: admin presses "Mark start of new user run"

#shot=19&t=777600&panel=admin&scroll=usage: scenario 19 on day 9.

- **The report covers** the 60 days ending at the end of the latest user run: helium dispensed, litres during runs, cylinders fitted and cylinder-equivalents.
- **Run #1** is user A across a 48 h beam downtime: one run, 23 purges, 4775 L.
- **Run #2** is user B, still in progress (2 purges so far). "Cylinders fitted 1" is the cylinder fitted at the start; user B's cylinder change comes on day 9.5.
- **Mark start of new user run** splits runs at a changeover.
- **Live internals** (below) list each forecast window's usage rate and run-out time.

12. Plant model parameters (simulator only)



#shot=1&t=60&details=1&scroll=plantGrid

- The physics behind the simulation:
 - volume and leak laws per lid
 - flow-off leak-in, cracked-lid leak, lid-lift time constant, open-lid air exchange
 - sensor-zone lag and transport delay
 - analyzer noise and the slow reading component
 - Alicat response, and the cylinder
- The controller never sees these.** Change them to test robustness, for example lower the open-lid air exchange to see where the lid check stops catching an open box.

13. Help

The screenshot shows the '15LSS_sample_gas: helium purge / O2 feedback simulator' interface. The main window displays various controls and readouts, including 'Oxygen 14.022', 'Flow setpoint 20.00', and 'State PURGE'. A 'Help' window is open, titled 'Sample-gas controller simulator: user guide'. The help window contains the following text:

Sample-gas controller simulator: user guide

simulator/sample_gas_simulator.html. It is a single file: open it in any browser (Edge, Chrome, Firefox) with nothing to install. Hover over almost anything to see a tooltip. ? Help (top right) shows this guide, the guide for building a similar simulator, and the self-test.

What it is, and what it is not

- **It is** an executable version of the design spec (docs/superpowers/specs/2026-09-24-o2-purge-feedback-design.md). The helium purge and O2 feedback controller for the 15LSS_sample_gas IOC runs against a model of the real enclosure, O2 analyzer, Alicat MFC and helium cylinder.
- **Its purpose** is to let you try the design before any EPICS code exists: press buttons, break things, and watch what the controller does.
- **It is not** the IOC. The real controller will be an epid record plus an SNL state machine. Anything you change here should go back into the spec, and the simulator's scenarios become the tests the real IOC must pass.
- **The physics comes from your data** (spec §2.1):
 - enclosure volume 41 L
 - leak rates per lid, and ~80 s dead time at hold flow
 - lid-lift kinetics
 - the Alicat's ramp and hold behaviour
 - the 1 Hz analyzer noise (white 0.84 m% plus a slow 0.67 m% component that is not bulk O2)

Quick start

1. Open the file. Station **15IDC** is selected: it starts IDLE, lid on, box full of air.
2. Press **Purge**. Watch the O2 graph fall on the log axis, and the flow jump to 20 SLPM.
3. Set **Speed** to 300× or 1000× to see the handoff to the PID and the regulation at 0.99 %.
4. In the right-hand column, press **Lift lid**. Within ~10 s the controller stops the flow (state OPEN_STOP, red banner).
5. Press **Close lid**, then **Purge** again.

To try a prepared case, pick a **Scenario** at the bottom right and press **Reset station & run scenario**. Each scenario's description says what to watch for.

Screen layout

Area	What it is
Header	Station tabs (15IDC, 15IDE; the dot shows the worst alarm), Pause/Run, +10 s, Speed, simulated clock, ? Help
Left column: main panel	What the operator's Phoebus main panel will show: alarm banner, the four big readouts, state, controls, buttons, Admin
Centre: graphs	O2, helium flow, helium left in the cylinder, and a strip showing the controller state over time
Right column: physical world	Simulator only. Change the lid, the Alicat, the cylinder, the analyzer, crash the IOC, run scenarios, edit the plant physics

#shot=1&t=60&panel=help

- **User guide:** the text version of this tour, with full detail.
- **Guide for building a similar simulator:** for an AI agent writing one for another loop, e.g. temperature control.
- **Self-test.**

14. Self-test

The screenshot displays the '15LSS_sample_gas: helium purge / O2 feedback simulator' interface. A 'Self-test' dialog box is open, explaining that the test runs every scenario on a hidden station with default settings and checks for correct termination. It includes a 'Run self-test' button and tabs for 'Scenario', 'Simulated', and 'Result'. The background interface shows various controls and data:

- Top Bar:** '15LSS_sample_gas: helium purge / O2 feedback simulator', 'executable version of the 2026-09-24 design spec, not IOC code', status indicators for '15IDC PURGE' and '15SIDE IDLE', 'Run' button, '+10 s' speed, 'Speed 60x', '0:01:00' timer, and a 'Help' button.
- Left Panel:** 'Sample gas 15IDC: SampleGas', 'No alarms', 'Oxygen 14.022 % O2 target 0.99 ± 0.020 %', 'Flow setpoint 20.00 SLPM (Setpoint_RBV) commanded 20.000', 'State PURGE in range', 'Purging at full flow until O2 < target - Δ (54.2)', 'purge 0:00:48 / 0:00:31 (scaled) start 19.21 % lid check: decay 96 % of lid-on lag-corrected 12.763 %, handoff at < 1.030 %', 'Enclosure mode A: Normal lid', 'O2 target (%) 0.99', 'Purge', 'Flow Zero', 'Resume PID', 'Last action: PURGE: purge started', 'New He cylinder fitted', 'Admin settings...'
- Center Panel:** 'Helium flow (SLPM)' graph showing 'Flow_RBV', 'commanded setpoint', and 'PID max (DRVH)'. Below it, 'Helium left in cylinder (L from MFC totalizer)' graph showing 'litres left'.
- Right Panel:** 'Alicat MFC' controls (Hold valve, Stuck hold, Clear hold, Ramp 3 SLPM/s, Set Gas He), 'Helium cylinder' controls (Refill, Nearly empty, Empty, cylinder pressure PV exists), 'O2 analyzer' (normal, frozen value, INVALID, Breath dip), 'Controller IOC' (Crash IOC, Restart IOC), 'Scenarios (spec §8)' (1. Normal purge from air, normal lid (mode A)), 'Reset station & run scenario', 'Reset station', and a note: 'Lid closed, box full of air. Purge at t = 5 s. Watch: lid check passes (decay ≈ 100 % of the lid-on rate). Handoff when the lag-corrected O2 < target - Δ for 5 s; the purge timeout scales with the target (≈ 1.3 ×)'. A 'Help' button is also present.
- Bottom Panel:** 'Event log' showing a list of events with timestamps and descriptions, including 'show both stations', 'MINOR = yellow, MAJOR = red. Times are the station's simulated clock.', and various status changes.

#shot=1&t=60&panel=selftest

- **Run self-test** runs every scenario on a hidden station with the **default** settings. It checks the end state and the key log messages, and takes about 10–30 s.
- **Your stations and settings are untouched.** A failing row states exactly what differed.
- Run it after any change to the simulator.